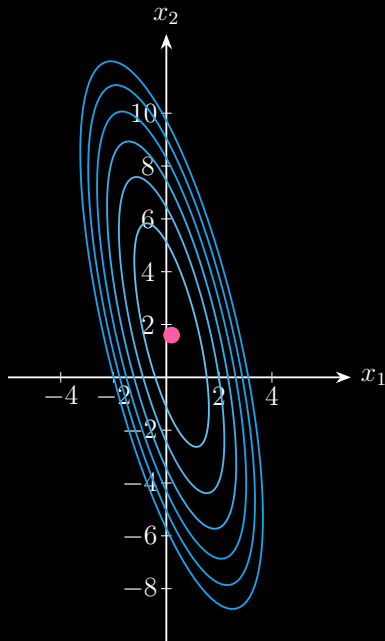


Quadratic optimization:

$$\min_x \quad x^\top A x - b^\top x$$

with

$$A = \begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$



Quadratic optimization:

$$\min_x \quad x^\top A x - b^\top x$$

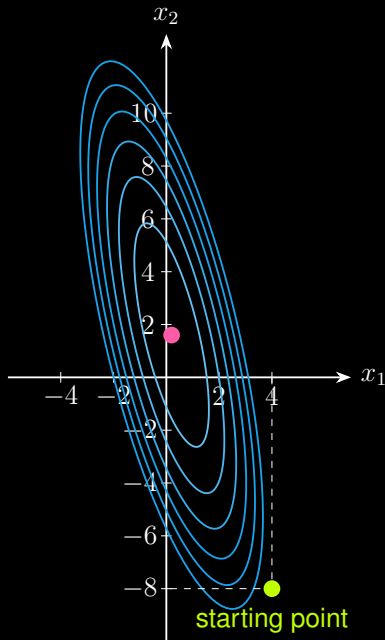
Steepest gradient descent:

$$\mathbf{g}_k = A\mathbf{x}_k - \mathbf{b} \quad (\text{gradient})$$

$$\alpha_k = \frac{\mathbf{g}_k^\top \mathbf{g}_k}{\mathbf{g}_k^\top A \mathbf{g}_k} \quad (\text{step size})$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \mathbf{g}_k \quad (\text{decision vector})$$

with an initial point $\mathbf{x}_0$.



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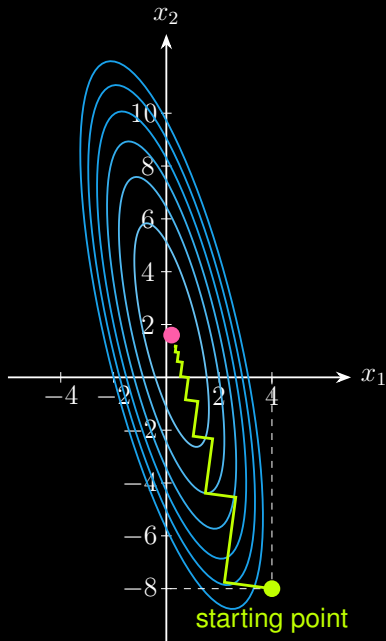
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Positive Definite Matrix $\mathbf{A} = \begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}$

Eigenvalue decomposition:

$$\mathbf{A} = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 9.47 \quad \lambda_2 = 0.53$$

and condition number:

$$\kappa = \frac{\lambda_1}{\lambda_2} = 17.94$$

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```
1 import numpy as np
2
3 A = np.array([[9, 2], [2, 1]])
4 eig_val, eig_vec = np.linalg.eig(A)
5
6 print('lambda_1 = ', eig_val[0])
7 print('lambda_2 = ', eig_val[1])
8 print('u_1 = ', eig_vec[:, 0])
9 print('u_2 = ', eig_vec[:, 1])
10 print('condition number = ',
11       eig_val[0] / eig_val[1])
```


Thanks for your attention!

About me:

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